TEXORA · LECTURE SHEET

Commercial Polymers

The polymers that industry actually makes — their monomers, repeat units, defining properties and uses — with the fibre-forming four (polyester, nylons, acrylic, polyolefins) treated in depth.

TE-2107 · POLYMER SCIENCE AND ENGINEERING · TOPIC 3 · LEVEL 2 TERM 1

1. The polyolefins: polyethylene and polypropylene

1.1 Polyethylene (PE)

Monomer ethylene, $CH_2{=}CH_2$ (28.05 g/mol); repeat unit $–[CH_2–CH_2]–$. Two commercial families:

- **LDPE** (low-density, ≈ 0.92 g/cm³) — made by high-pressure free-radical polymerisation; heavily branched, soft, flexible, melts about 105–115 °C. Film, bags, coatings; too weak and low-melting for apparel fibre.
- **HDPE** (≈ 0.95–0.97 g/cm³) — made with Ziegler–Natta or Phillips catalysts at low pressure; almost linear, highly crystalline, stiff, melts about 130–137 °C. Ropes, nets, geotextiles, bottles and crates.

PE has only van der Waals cohesion: it cannot be dyed by normal methods, floats on water, and is chemically almost inert.

1.2 Polypropylene (PP)

Monomer propylene, $CH_2{=}CH–CH_3$; repeat unit $–[CH_2–CH(CH_3)]–$. Only the **isotactic** form (Topic 1) crystallises into a useful material: density 0.90–0.91 g/cm³ (the lightest common fibre), melting point about 165 °C, moisture regain essentially zero, excellent chemical resistance. Because it has no dye sites, PP is coloured by pigment mixed into the melt before spinning (**solution-dyed / dope-dyed**). Uses: carpets and carpet backing, upholstery, ropes, nonwoven hygiene products, sacks, and thermal underwear (it transports liquid sweat without absorbing it).

2. Poly(ethylene terephthalate) — PET polyester

The world's dominant fibre polymer. Two monomers: **terephthalic acid** (TPA, 166.13 g/mol), or its ester dimethyl terephthalate (DMT), and **ethylene glycol** (EG, 62.07 g/mol). In the DMT route the first stage is transesterification to bis(2-hydroxyethyl) terephthalate (BHET) with methanol removed; in the

TPA route, direct esterification with water removed. The second stage is melt polycondensation under high vacuum at about 280 °C, glycol being distilled off until the melt reaches fibre-grade molar mass:

$$\text{n HOOC–C}_6\text{H}_4\text{–COOH + n HO–CH}_2\text{CH}_2\text{–OH} \rightarrow \text{–[CO–C}_6\text{H}_4\text{–CO–O–CH}_2\text{CH}_2\text{–O]}_n\text{–} + 2n\ \text{H}_2\text{O}$$

Repeat unit $C_{10}H_8O_4$ = 192.17 g/mol. The stiff benzene ring in the backbone gives PET its high melting point (≈ 255–260 °C), high modulus and excellent crease recovery; the ester groups give moderate polarity. Moisture regain only about 0.4 %, so it dries fast but builds static and needs disperse dyes (no ionic sites). Density 1.38 g/cm³. Uses: apparel (alone and blended with cotton), filament and staple, bottles (higher molar mass grade), tyre cord, conveyor belts, film.

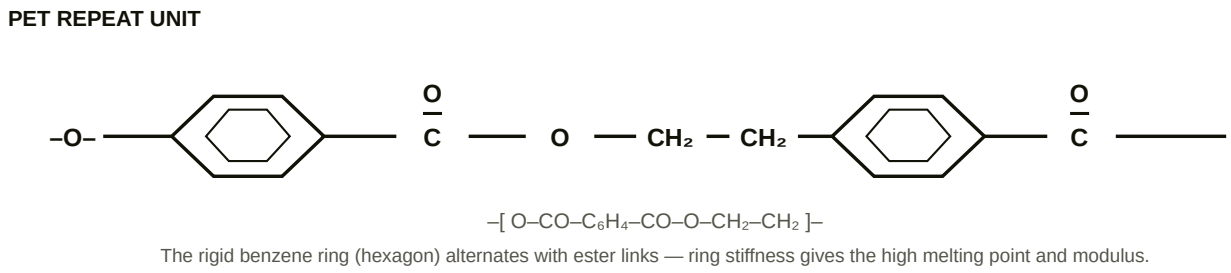


Figure 1 — Structural schematic of the PET repeat unit (not to scale; ring drawn as hexagon, double bonds of the ring and C=O indicated by the inner hexagon and the O above each carbonyl carbon).

3. The nylons (polyamides)

3.1 Nylon 6,6

From **hexamethylenediamine** $H_2N–(CH_2)_6–NH_2$ (116.21 g/mol) and **adipic acid** $HOOC–(CH_2)_4–COOH$ (146.14 g/mol). The two are first combined in exact 1:1 ratio as "nylon salt" — the built-in stoichiometry guarantee that Carothers demands (Topic 2) — then melt-polymerised at about 270–280 °C with water removed:

$$\text{n H}_2\text{N–(CH}_2\text{)}_6\text{–NH}_2 + \text{n HOOC–(CH}_2\text{)}_4\text{–COOH} \rightarrow \text{–[NH–(CH}_2\text{)}_6\text{–NH–CO–(CH}_2\text{)}_4\text{–CO]}_n\text{–} + 2n\ \text{H}_2\text{O}$$

Repeat unit = 226.32 g/mol (Topic 1). The amide (–CONH–) groups hydrogen-bond chain to chain, giving melting point ≈ 255–265 °C, high strength, outstanding abrasion resistance and toughness; regain about 4 % lets acid dyes attach to the amine end groups.

3.2 Nylon 6

From the cyclic monomer **caprolactam** (113.16 g/mol) by ring-opening polymerisation with a trace of water at about 250–270 °C; the repeat unit –[NH–(CH_2)$_5$–CO]– has the same composition as one nylon 6,6 "half". Melting point lower, ≈ 215–220 °C; properties otherwise similar, slightly softer and easier to dye. The "6" names the carbons in caprolactam; "6,6" counts the carbons of diamine and diacid — the numbering system identifies every polyamide.

FEATURE	NYLON 6	NYLON 6,6
Monomer(s)	Caprolactam (ring-opening)	Hexamethylenediamine + adipic acid
Melting point	≈ 215–220 °C	≈ 255–265 °C
Hand feel	Softer	Slightly harsher, springier
Typical uses	Hosiery, carpets, apparel	Tyre cord, hosiery, engineering parts

4. Acrylic (polyacrylonitrile, PAN)

Monomer **acrylonitrile** CH_2=CH–C≡N (53.06 g/mol), polymerised by free-radical chain growth, almost always as a **copolymer**: fibre-grade acrylic contains at least 85 % acrylonitrile with a few percent of a neutral comonomer (e.g. methyl acrylate) to loosen the structure and a trace of an ionic comonomer to provide dye sites. The dense thicket of strongly polar nitrile groups gives high inter-chain forces — PAN does not melt; it decomposes above about 300 °C, so acrylic is solution-spun, never melt-spun (Topic 5 of TE-2108). Properties: wool-like bulk and warmth, excellent sunlight resistance (best of the common fibres), regain about 1.5–2 %. Uses: knitwear, blankets, awnings, and as the precursor of carbon fibre.

5. Other commercial polymers

POLYMER	MONOMER / ROUTE	KEY PROPERTIES	MAIN USES
PVC	Vinyl chloride, suspension polymerisation	Unplasticised: rigid, flame-retardant; plasticised: flexible	Pipes, cable insulation, coated fabrics, synthetic leather, rainwear
Polystyrene	Styrene, bulk/suspension	Glassy, brittle, clear; expanded form (EPS) is foam	Packaging, insulation foam, disposable articles
PMMA (acrylic glass)	Methyl methacrylate, bulk casting	Outstanding clarity, weather-resistant	Sheets, lenses, displays — not a textile fibre

Polyurethane / spandex	Diisocyanate + polyol (step growth, block copolymer)	Elastomeric; 400–700 % extension with recovery	Stretch fabrics, foam, coatings, synthetic leather
PVA	Vinyl acetate polymerised, then hydrolysed	Water-soluble grades exist; strong films	Vinylon fibre, adhesives, warp sizing (later washed out)
PTFE	Tetrafluoroethylene	Melts ≈ 327 °C, almost zero friction, totally inert	Non-stick coatings, breathable membranes (expanded PTFE), sewing threads
Aramids	Aromatic diamines + diacid chlorides (solution polycondensation)	Do not melt; extreme strength (para-aramid) and heat resistance (meta-aramid)	Protective clothing, ballistic fabrics, composites — detailed in TE-2108 Topic 4

WORKED EXAMPLE — WHICH POLYMER, WHICH PROCESS?

A mill needs a fibre that (i) survives ironing at 200 °C, (ii) can be dyed with disperse dyes, (iii) has regain below 1 %.

Check the candidates: nylon 6 melts ≈ 220 °C — ironing margin too small, regain 4 % fails (iii). Acrylic cannot be melt-processed and degrades with heat; regain ≈ 2 % fails (iii). Polypropylene melts ≈ 165 °C — fails (i). **PET**: melts ≈ 255–260 °C ✓, disperse-dyed ✓, regain ≈ 0.4 % ✓ — polyester is the answer, and the benzene-ring stiffness behind its melting point is exactly what Figure 1 shows.

Glossary

Nylon salt — the 1:1 crystalline salt of hexamethylenediamine and adipic acid, guaranteeing stoichiometric balance. **Dope-dyed** — pigment dispersed in the polymer melt/solution before spinning, so colour runs through the whole filament. **Ziegler–Natta catalyst** — a coordination catalyst system producing linear, stereoregular polyolefins at low pressure. **Transesterification** — exchange of alcohol components of an ester, the first stage of the DMT route to PET. **BHET** — bis(2-hydroxyethyl) terephthalate, the PET prepolymer intermediate.

Key points

• PET (192.17 g/mol repeat unit): benzene-ring stiffness → high melting point, modulus and crease recovery; regain only ≈ 0.4 %, disperse dyes only.

• Nylon 6,6 (226.32 g/mol): amide hydrogen bonding gives strength, abrasion resistance and a ≈ 260 °C

melting point; nylon 6 from caprolactam melts ≈ 40 °C lower.

• PP is the lightest fibre polymer (0.90 g/cm³) and must be dope-dyed; PE splits into branched LDPE and linear HDPE.

• Acrylic is a PAN copolymer (≥ 85 % acrylonitrile); nitrile polarity prevents melting, so it is solution-spun.

• Commercial selection is property-driven: melting point, regain, dyeability and density decide the polymer before price is discussed.

Practice

1. Write the polymerisation equation for nylon 6,6 and show where the number 226.32 g/mol for its repeat unit comes from.
2. Why is the nylon-salt stage essential for nylon 6,6 but unnecessary for nylon 6?
3. Polypropylene garments are never piece-dyed in a normal dyehouse range. Explain why, and state how their colour is obtained.
4. Calculate the DP of a PET sample of molar mass 24,977 g/mol.
5. Give two structural reasons why PET's melting point (≈ 258 °C) is so much higher than HDPE's (≈ 135 °C).

Answer key

1. n $H_2N(CH_2)_6NH_2$ + n $HOOC(CH_2)_4COOH$ → $–[NH(CH_2)_6NH–CO(CH_2)_4CO]_n–$ + 2n H_2O. Repeat unit = 116.21 + 146.14 − 2(18.02) = **226.32 g/mol**.
2. Nylon 6,6 has two monomers; step growth reaches fibre DP only at exact 1:1 balance (Carothers), which the salt's fixed composition guarantees. Nylon 6 polymerises from a single monomer by ring opening, so there is no ratio to get wrong.
3. PP is a pure hydrocarbon with no polar or ionic dye sites and essentially zero regain — dye molecules have neither attraction nor entry path. Colour is added as pigment in the melt before extrusion (dope dyeing).
4. DP = 24,977 ÷ 192.17 = **130**.
5. (i) The benzene ring makes the PET backbone stiff, so chains lose less entropy on melting and need more thermal energy to disorder. (ii) The polar ester groups add dipole attractions between chains, whereas HDPE relies on weak van der Waals forces alone.